Origins of Mathematics — Problem Set 0

Answer Key

Show your work and explain your reasoning. Use the notation introduced in the relevant passus.

1 A simple grouping system

The symbols represent

$$\bullet = 1, \qquad \blacktriangle = 4, \qquad \blacksquare = 4^2 = 16, \qquad \blacklozenge = 4^3 = 64.$$

This is a base-4 simple grouping system (each symbol may be repeated up to three times).

(a) $\blacklozenge\blacklozenge\blacksquare\bullet = 2(64) + 1(16) + 1(1) = 128 + 16 + 1 = \boxed{145}$.

(b) $\blacklozenge\blacksquare\blacksquare\blacksquare\blacktriangle = 64 + 3(16) + 4 = 64 + 48 + 4 = \boxed{116}$.

(c) $81 = 64 + 16 + 1 = \blacklozenge\blacksquare\bullet$.

(d) $210 = 3(64) + 1(16) + 0(4) + 2(1) = 192 + 16 + 4 + 2$, so

$$210 = \blacklozenge\blacklozenge\blacklozenge\blacksquare\blacktriangle\bullet\bullet.$$

(Check: $3 \cdot 64 = 192$, remainder 18; $1 \cdot 16 = 16$, remainder 2; $0 \cdot 4$; $2 \cdot 1 = 2$.)

2 Inventing a base-11 system

Sample solution. Choose eleven distinct symbols:

value	0	1	2	3	4	5	6	7	8	9	10
symbol	$\odot$	$\star$	$\triangle$	$\square$	$\diamond$	$\heartsuit$	$\clubsuit$	$\spadesuit$	$\oplus$	$\otimes$	θ

(a) Three examples (subscript 11 denotes base 11):

$$\star\odot\star_{11} = 1 \cdot 11^2 + 0 \cdot 11 + 1 = 121 + 1 = 122_{10},$$
$$\theta\triangle_{11} = 10 \cdot 11 + 2 = 112_{10},$$
$$\star\star\star_{11} = 1 \cdot 121 + 1 \cdot 11 + 1 = 133_{10}.$$

(b) Take $\theta\star\triangle_{11}$ (three different symbols):

$$\theta\star\triangle_{11} = 10 \cdot 121 + 1 \cdot 11 + 2 = 1210 + 11 + 2 = 1223_{10}.$$

Convert to base 5:

$$1223 = 5 \cdot 244 + 3, \quad 244 = 5 \cdot 48 + 4, \quad 48 = 5 \cdot 9 + 3, \quad 9 = 5 \cdot 1 + 4, \quad 1 = 5 \cdot 0 + 1.$$

Hence

$$1223_{10} = \boxed{14343_5}.$$

3 Multiplication in base 6

Products $a \cdot b$ for $a, b \in \{1, 2, 3, 4, 5, 10\}$, all written in base 6:

$\cdot$	1	2	3	4	5	10
1	1	2	3	4	5	10
2	2	4	10	12	14	20
3	3	10	13	20	23	30
4	4	12	20	24	32	40
5	5	14	23	32	41	50
10	10	20	30	40	50	100

Sample checks: $4 \cdot 5 = 20_{10} = 32_6$ (since $3 \cdot 6 + 2 = 20$); $5 \cdot 5 = 25_{10} = 41_6$ ($4 \cdot 6 + 1 = 25$); $10 \cdot 10 = 100_{10} = 100_6$ ($1 \cdot 36 = 36$).

4 Babylonian notation

(a) The sequence "horizontal wedge, vertical wedge, vertical wedge, vertical wedge" reads as the numeral 13 (one ten plus three ones) in a single place, but because spacing and zero were ambiguous it admits multiple readings:

1. Single place: 13_{10}.
2. Two places, $13; 0$: $13 \cdot 60 = 780_{10}$.
3. Two places, $0; 13$: $13/60 \approx 0.2167_{10}$.
4. Two places, $1; 3$ (splitting the ten-wedge from the units): $1 \cdot 60 + 3 = 63_{10}$.
5. Three places, $13; 0; 0$: $13 \cdot 60^2 = 46800_{10}$.

(Other readings are possible; any four valid, clearly explained interpretations earn full credit.)

(b) $120\frac{1}{3} = 2 \cdot 60 + 0 + \frac{20}{60}$. Therefore

$$120\tfrac{1}{3} = \boxed{2; 0; 20}$$

where the semicolon marks the sexagesimal point (so this is $2 \cdot 60 + 0 + 20/60 = 120 + \frac{1}{3}$).

5 Magic squares

(a) The completed normal 4×4 magic square (magic constant 34):

1	14	4	15
8	11	5	10
13	2	16	3
12	7	9	6

Check: rows $1+14+4+15 = 34$, $8+11+5+10 = 34$, $13+2+16+3 = 34$, $12+7+9+6 = 34$; columns $1+8+13+12 = 34$, $14+11+2+7 = 34$, $4+5+16+9 = 34$, $15+10+3+6 = 34$; diagonals $1+11+16+6 = 34$, $15+5+2+12 = 34$. All integers 1–16 are used exactly once.

(b) (Extra credit.) The entries sum to

$$1 + 2 + \cdots + n^2 = \frac{n^2(n^2+1)}{2}.$$

Since there are n rows, each with the same sum S,

$$nS = \frac{n^2(n^2+1)}{2} \quad \Longrightarrow \quad \boxed{S = \frac{n(n^2+1)}{2}}.$$

For $n = 4$: $S = \frac{4 \cdot 17}{2} = 34$.

6 Unit fractions

Identity. For $n \geq 1$,

$$\frac{1}{n+1} + \frac{1}{n(n+1)} = \frac{n}{n(n+1)} + \frac{1}{n(n+1)} = \frac{n+1}{n(n+1)} = \frac{1}{n}.$$

Application. With $n = 101$:

$$\frac{1}{101} = \frac{1}{102} + \frac{1}{101 \cdot 102} = \boxed{\frac{1}{102} + \frac{1}{10302}}.$$

7 A second decomposition (extra credit)

Apply the identity from Problem 6 to $\frac{1}{101}$:

$$\frac{2}{101} = \frac{1}{101} + \left(\frac{1}{102} + \frac{1}{10302}\right).$$

The denominators $101, 102, 10302$ are distinct, so

$$\boxed{\frac{2}{101} = \frac{1}{101} + \frac{1}{102} + \frac{1}{10302}}.$$

(Check numerically: $\frac{1}{101} + \frac{1}{102} + \frac{1}{10302} \approx 0.009901 + 0.009804 + 0.0000971 = 0.019802 \approx \frac{2}{101}$. Many other valid answers exist.)

8 Adding fractions in *The Nine Chapters*

Let the fractions be $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \ldots, \frac{a_k}{b_k}$.

The procedure. "Multiply each numerator by the denominators of the other fractions" produces

$$N = a_1 b_2 b_3 \cdots b_k + a_2 b_1 b_3 \cdots b_k + \cdots + a_k b_1 b_2 \cdots b_{k-1}.$$

"Multiply the denominators to obtain the divisor" produces

$$D = b_1 b_2 \cdots b_k.$$

The result is $\dfrac{N}{D}$ (reduced if a remainder arises, per the final instruction).

Why it works. Each term $a_i \prod_{j \neq i} b_j$ equals $a_i \cdot \dfrac{D}{b_i}$, i.e. the i-th fraction rewritten over the common denominator D:

$$\frac{a_i}{b_i} = \frac{a_i \, (D/b_i)}{D}.$$

Summing the numerators over the common denominator D gives exactly the sum of the fractions:

$$\frac{N}{D} = \sum_{i=1}^{k} \frac{a_i}{b_i}.$$

The procedure is thus the standard "common denominator" algorithm, where the common denominator is taken to be the product of all denominators (not necessarily the least common multiple). The final instruction (remainder over divisor) simply expresses the result as a mixed number / proper fraction.

9 "Seven has no inverse"

In base-60 computation, the reciprocal of n is a *terminating* sexagesimal exactly when n's only prime factors are $2, 3, 5$ — because $60 = 2^2 \cdot 3 \cdot 5$. Since 7 is prime and does not divide any power of 60, $\frac{1}{7}$ has a purely repeating (non-terminating) sexagesimal expansion:

$$\frac{1}{7} = 0; \overline{8, 34, 17}_{60}$$

(since $\frac{1}{7} = \frac{8}{60} + \frac{34}{3600} + \frac{17}{216000} + \cdots$). Babylonian scribes, working with finite sexagesimal "tables of reciprocals," could not express $\frac{1}{7}$ exactly as a finite sequence of sexagesimal places. Saying "seven has no inverse" means: **7 has no finite reciprocal in base 60** — there is no finite sexagesimal number whose product with 7 is exactly 1. (Contrast: $2, 3, 4, 5, 6, 8, 9, 10, 12, 15, \ldots$ do have terminating reciprocals.)

10 A quantity from the Rhind Mathematical Papyrus

Let the quantity be x. Then

$$x + \frac{x}{3} + \frac{x}{4} = 2.$$

Combine:

$$x\left(1 + \frac{1}{3} + \frac{1}{4}\right) = x \cdot \frac{12 + 4 + 3}{12} = \frac{19x}{12} = 2.$$

Hence

$$x = \frac{24}{19} = 1 + \frac{5}{19}.$$

Answer: $\boxed{\dfrac{24}{19}}$, i.e. $1\frac{5}{19}$.

Egyptian-style check (method of false position): guess $x = 12$; then $12 + 4 + 3 = 19$, but we want 2. Scale by $2/19$: $x = 12 \cdot \frac{2}{19} = \frac{24}{19}$. In unit fractions, $\frac{24}{19} = 1 + \frac{1}{4} + \frac{1}{76}$, since $\frac{5}{19} = \frac{20}{76} = \frac{19}{76} + \frac{1}{76} = \frac{1}{4} + \frac{1}{76}$.

11 Egyptian multiplication

(a) Compute 26×31 by doubling:

	doubling of 26	accumulate?
1	26	✓
2	52	✓
4	104	✓
8	208	✓
16	416	✓

Since $31 = 1 + 2 + 4 + 8 + 16$,

$$26 + 52 + 104 + 208 + 416 = 806.$$

So $26 \times 31 = \boxed{806}$.

(b) Example: 23×45. Doublings of 23:

$$1 \to 23, \quad 2 \to 46, \quad 4 \to 92, \quad 8 \to 184, \quad 16 \to 368, \quad 32 \to 736.$$

Since $45 = 32 + 8 + 4 + 1$,

$$736 + 184 + 92 + 23 = 1035.$$

So $23 \times 45 = \boxed{1035}$.

(c) Verification:

$$26 \cdot 31 = 26 \cdot 30 + 26 = 780 + 26 = 806, \qquad 23 \cdot 45 = 23 \cdot 40 + 23 \cdot 5 = 920 + 115 = 1035.$$

(d) **Why it always works.** The Egyptian method writes the multiplier b in binary: $b = \sum_{i \in S} 2^i$ for some set S of exponents (every positive integer has a unique binary expansion). The table computes $a \cdot 2^i$ for each i by repeated doubling, using the distributive law:

$$a \cdot b = a \cdot \sum_{i \in S} 2^i = \sum_{i \in S} a \cdot 2^i.$$

Adding the selected table entries therefore gives exactly $a \cdot b$. Since every positive integer has a binary representation, the method terminates and is correct for all positive integers a, b.

12 Egyptian division

Compute $132 \div 6$ by asking: what times 6 gives 132? Build doublings of 6:

multiple	value
1	6
2	12
4	24
8	48
16	96

Now $132 = 96 + 36$, and $36 = 24 + 12$, and $12 = 12$. Thus

$$132 = 96 + 24 + 12 = 6 \cdot 16 + 6 \cdot 4 + 6 \cdot 2 = 6 \cdot (16 + 4 + 2) = 6 \cdot 22.$$

Therefore

$$132 \div 6 = \boxed{22}.$$

(Verification: $6 \cdot 22 = 132$.)

13 Empty places on a Chinese counting board

(a) On a counting board, each decimal place is a column (or row), and the value in each place is shown by rods. An **empty position** denotes a zero in that place.

- **205:** rods for 2 in the hundreds place, *nothing* in the tens place, rods for 5 in the units place:

2	(empty)	5

- **25:** rods for 2 in the tens place, rods for 5 in the units place; the hundreds place is simply absent (only two occupied positions):

(empty)	2	5

The empty tens position in 205 is a *deliberate* gap between the 2 and the 5; in 25 there is no gap — the 2 and 5 occupy adjacent places.

(b) **Comparison with Babylonian notation.** On a counting board the *physical column structure* is fixed and visible: each column has a definite place value (units, tens, hundreds, …), so an empty column unambiguously means "zero in this place," and its position relative to neighboring columns is manifest. In Babylonian cuneiform, by contrast, numerals were written as a linear string of wedges on clay with no fixed columns; the scribe had to rely on spacing to separate sexagesimal places, and a missing place (zero) was often simply not written. The reader could not tell whether a gap meant "zero here" or merely a wide space between adjacent places — hence the ambiguity of Problem 4(a). The counting board's grid supplies the place structure that cuneiform writing lacked, making the empty position informative rather than ambiguous.